
GROUP 3 MODULE III PROJECT

Movie Industry Data Analysis

GROUP MEMBERS

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PROJECT DEFENSE

1. Business Problem & Objectives

Turning historical movie data into practical decision support.

Movie studios must balance release timing, genre selection and production investment.

The project combines financial movie data with IMDb information to identify performance patterns.

Objective: support better decisions about release windows, genre prioritization and production budgets.

Key measures: worldwide gross, production budget, ROI, release month, genre and audience rating.

2. Data Sources & Project Structure

Financial files + IMDb SQLite data + documented project workflow.

Movie Budget Data

Production budget, domestic gross, worldwide gross, release date and movie title.

IMDb Database

Movie titles, genres, average ratings and vote counts. IMDb tables are connected through movie_id.

Project Assets

zippedData/, im.db, ERD image, Jupyter notebook, README and supporting documentation.

3. Data Preparation & Feature Engineering

Preparing raw data for reliable analysis.

Removed '\$' and commas from monetary fields and converted them to numeric values.

Converted release_date to datetime and extracted release_month.

Calculated net_profit = worldwide_gross – production_budget.

Calculated ROI = (net_profit / production_budget) × 100.

Created loss_or_profit to classify movies as Profit or Loss.

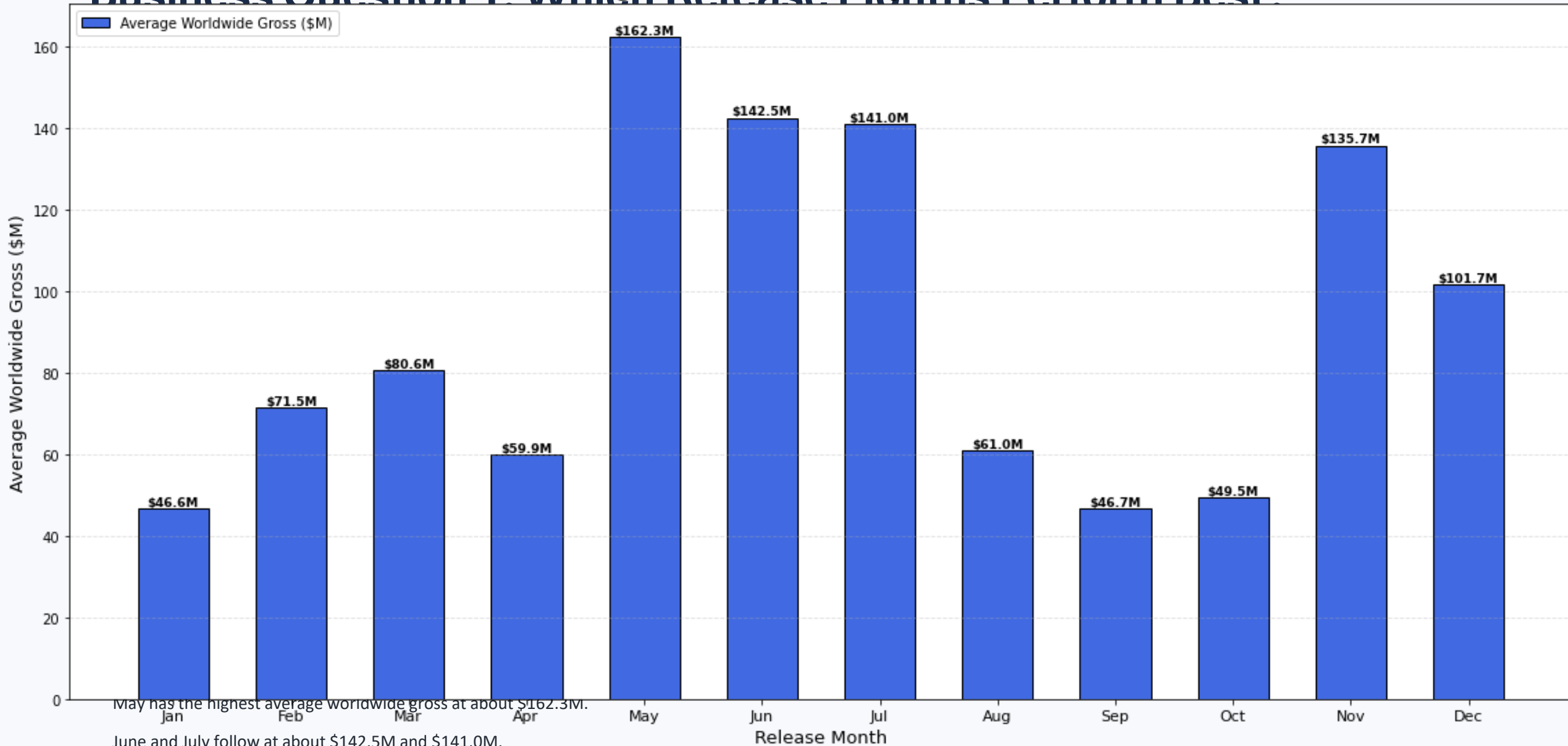
Split multi-genre strings and used explode() to analyze individual genres.

4. Analytical Workflow

From raw data to business recommendations.



Business Question 1: Which Release Months Perform Best?



December has the highest release volume (745), showing that volume and gross performance are different measures.

Business Question 1: Conclusion & Recommendation

Conclusion

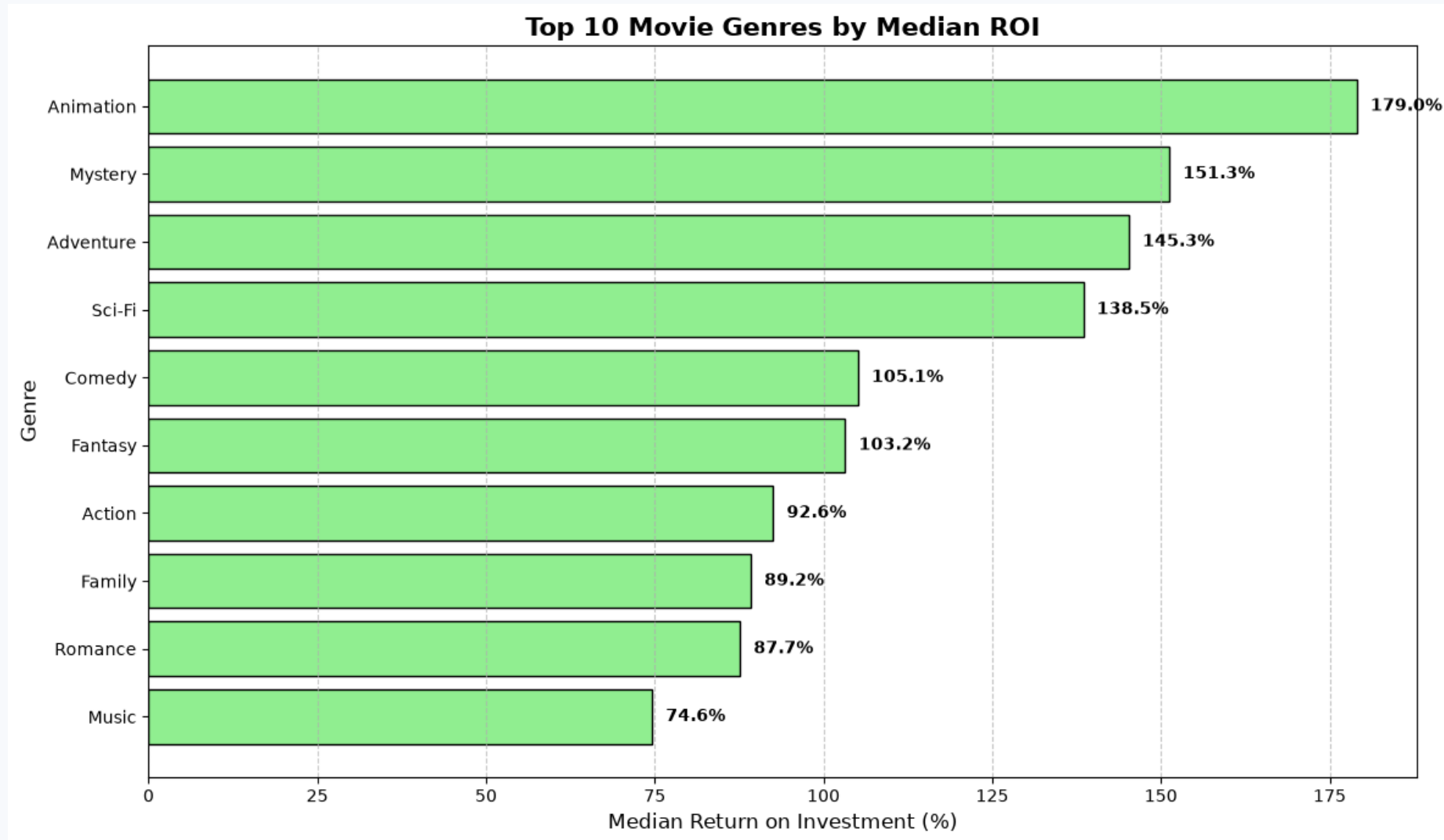
May–July and November–December emerge as prime release windows. May is the strongest individual month by average worldwide gross.

Recommendation

Prioritize commercially strong releases in these windows while considering competition, genre, marketing, audience behavior and release volume.

Business Question 2: Which Genres Have the Highest ROI?

Top 10 genres ranked by median ROI.



Business Question 2: Conclusion & Recommendation

Conclusion

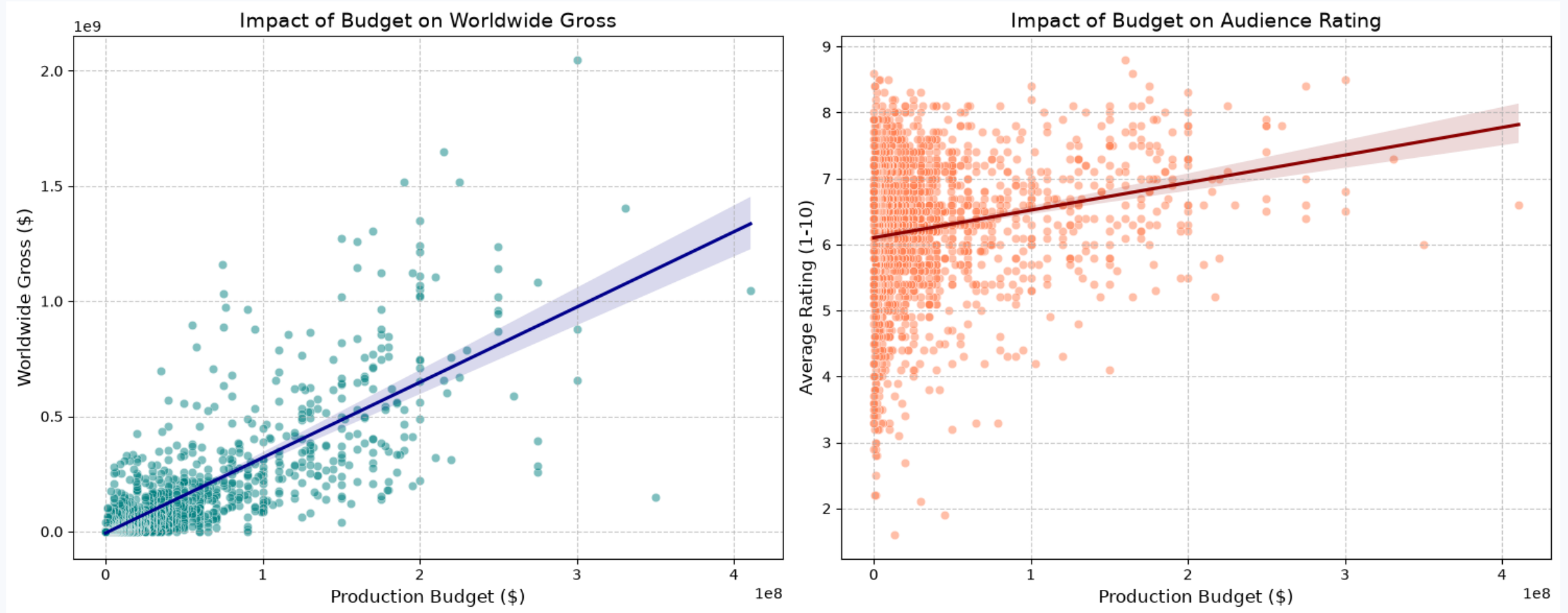
Animation leads with a 179.0% median ROI, followed by Mystery (151.3%), Adventure (145.3%) and Sci-Fi (138.5%).

Recommendation

Give greater consideration to high-ROI genres, but combine genre with budget, target audience, competition, release timing and marketing strategy.

Business Question 3: How Does Production Budget Affect Performance?

Budget is compared with worldwide gross and audience rating.



Higher production budgets generally correspond with higher worldwide gross.

The relationship is not perfect: similar budgets can produce very different revenues.

Production budget shows a relatively weak relationship with audience rating.

Business Question 3: Conclusion & Recommendation

Conclusion

Higher budgets are generally associated with higher worldwide gross, but budget size alone does not guarantee commercial success or higher audience ratings.

Recommendation

Optimize production budgets using expected revenue, historical ROI, genre performance, release timing, target audience, competition and marketing requirements.

Overall Findings

The three business questions form one decision framework.

WHEN?

May–July and November–December are the prime release windows identified by historical average worldwide gross.

WHAT?

Animation, Mystery, Adventure and Sci-Fi show the strongest median ROI among the top 10 genres.

HOW MUCH?

Optimize production budgets based on expected return and risk rather than assuming bigger is always better.

Genre → Budget → Expected ROI → Release Window → Marketing Strategy → Investment Decision

Limitations & Future Work

What the analysis can—and cannot—tell us.

Historical relationships do not guarantee future box-office outcomes.

ROI is based on worldwide gross and production budget; additional costs such as marketing are not included.

Multi-genre movies can contribute to more than one genre after explode().

Future work could incorporate marketing spend, competition, franchise strength, distribution strategy and audience demographics.

Predictive modeling could estimate expected gross or probability of profitability for future releases.

Final Takeaway

Movie profitability is influenced by a combination of

RELEASE TIMING • GENRE • BUDGET

Data-driven analysis can turn these factors into more informed investment and release decisions.

Thank You | Questions & Discussion